Compact Representation, Compression, Continuation, and Spectra of Response Functions in Time and Frequency

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Postdoc positions available!









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Phys. Rev. B 110, 035154 (2024)

Response functions

Quantum system

$\hbar$

Ground states, densities, band structure, entanglement



Optics, resistivities, scattering





Linear response





Finite-temperature, statistical mechanics, ensembles: finite-temperature field theories

Response functions

Green's functions

Susceptibilities

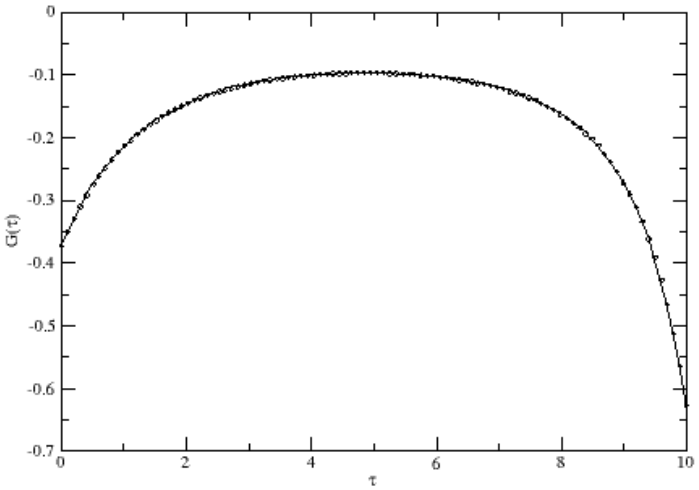
Propagators

Want to:
generate, store, calculate, analyze, …

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How to represent a function



Make a table!

```
0 −0.37407
0.10101 −0.350502
0.20202 −0.329075
0.30303 −0.309572
0.40404 −0.291798
0.505051 −0.275581
0.606061 −0.260766
0.707071 −0.247217
0.808081 −0.234811
0.909091 −0.223438
1.0101 −0.213002
1.11111 −0.203414
1.21212 −0.194597
1.31313 −0.18648
1.41414 −0.179001
1.51515 −0.172104
1.61616 −0.165737
1.71717 −0.159856
….
```

Can we do better?

- Find a better grid and **discretize adaptively**
- **Expand in orthogonal basis**, tabulate coefficients

Discretize uniformly

- What about values that aren't stored? (Interpolation, extrapolation)?
- What about doing Math?

How many points needed?

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How to represent a response function

Not much more we can do without knowing more about the function! But in many-body theory, many of the functions of interest are response functions. From here on out: we specialize on steady-state and equilibrium

- Response functions live in the **complex plane**
- In frequency space, response functions (real frequency, retarded, advanced, greater, lesser, …) are related to so-called **Nevanlinna functions**
- Nevanlinna functions have a **continued fraction expansion** (Schur algorithm)
- A continued fraction can be written as a **pole expansion**

$$G(z) = \sum_{l=1}^{M} \frac{A_l}{z - \xi_l},$$

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J. Chem. Phys. 162, 214111 (2025)

Representing response functions approximately

Approximate response functions with a compact **pole approximation**. For z on the real axis:

$$\left| G(z) - \sum_{l=1}^{M} \frac{A_l}{z - \xi_l} \right| \leq \varepsilon$$

M poles with pole residues A, pole locations xi, for approximation tolerance epsilon

Minmax theorem and analyticity ensure tolerance away from real axis

Minimum information principle: want most compact representation of a complex function. 4 real numbers for each pole (2 pole locations, 2 residues).

Q: **How many poles** do we need to represent a function to within a given precision?
Q: What are **efficient operations** we can do in this representation?
Q: **How** do we construct this pole approximation?

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Closely related: Moment representations

- On the real axis, response functions on are related to **probability distribution functions** (‘how many particles at which energy?’
- We can characterize probability distributions by their **moments** $m_k = \int_{\mathbb{R}} \omega^k A(\omega) d\omega,$
- If we only know only some of the moments, we can approximate the probability distribution: **Truncated Moment Problem** (Akhiezer 1965)
- For stability reasons: use a holomorphic mapping from the real axis to the unit circle and solve the so-called ***trigonometric*** **moment problem** $h_k = \int_{|z|=1} z^k \tilde{A}(z) dz,$
- Can readily compute moments from poles (residue theorem) and poles from the moments (ESPRIT)
- Moments are ideal for **machine learning & compression** and are **additive**

Dong, Gull, arXiv:2607.11190

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Q: how many poles do we need?

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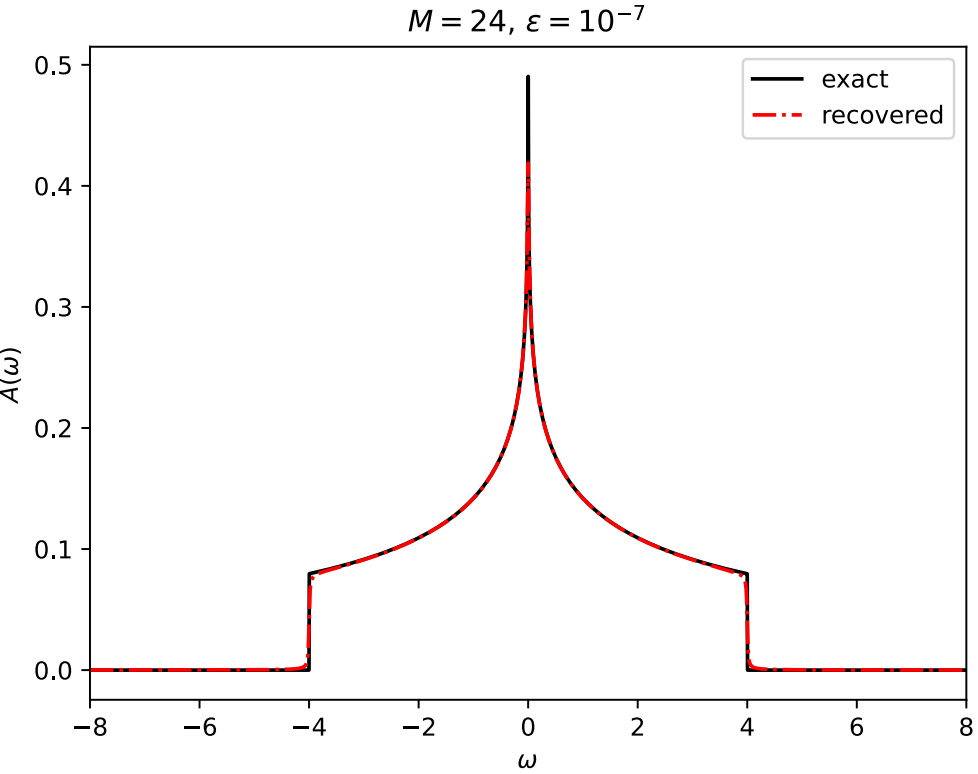
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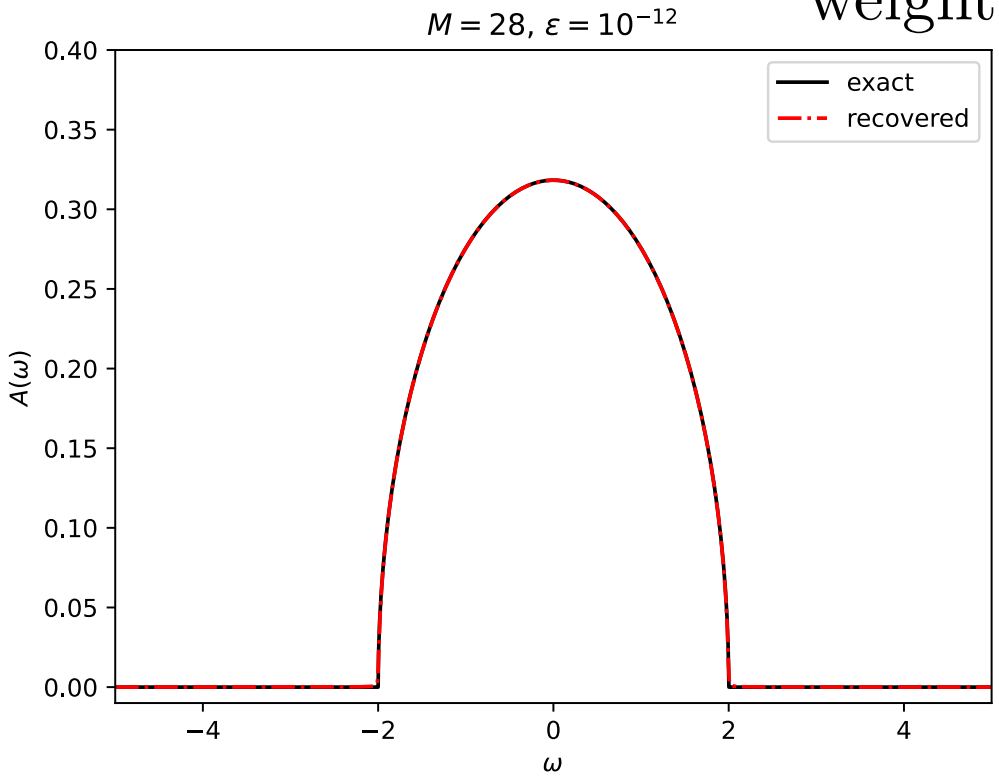
Pole approximations

$$G(z) = \sum_{l=1}^{M} \frac{A_l}{z - \xi_l},$$

$\xi_l \in \mathbb{C}$ denote M pole locations

$A_l \in \mathbb{C}$ the corresponding complex weights





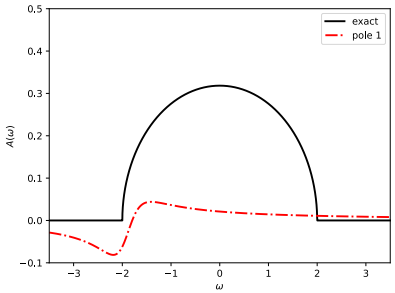
Pole approximations

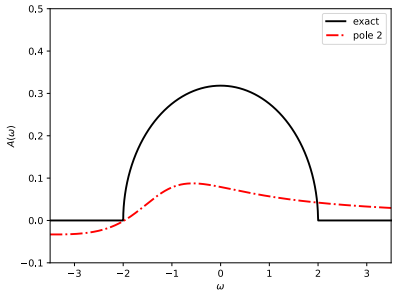
$$G(z) = \sum_{l=1}^{M} \frac{A_l}{z - \xi_l},$$

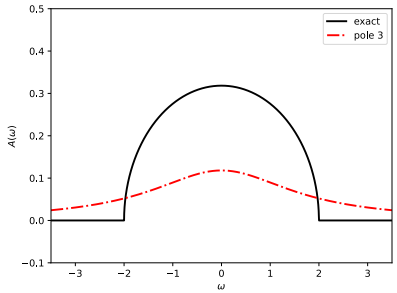
$\xi_l \in \mathbb{C}$ denote M pole locations

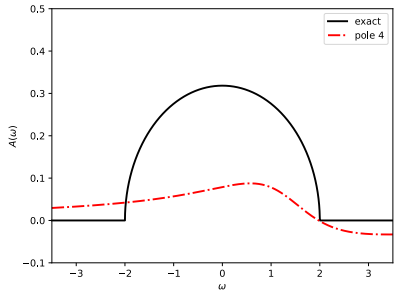
$A_l \in \mathbb{C}$ the corresponding complex weights

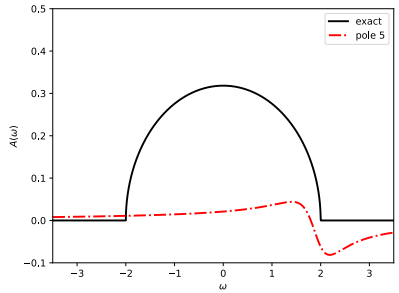
Note that the **residues** must be allowed to be **complex**

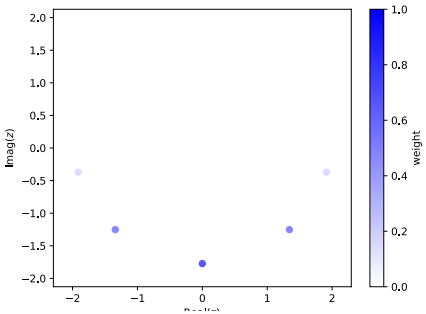












Individual poles that make up the pole approximation of the semicircular DOS with five components (accuracy 10**-2)

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Q: what can we do

efficiently

with poles?

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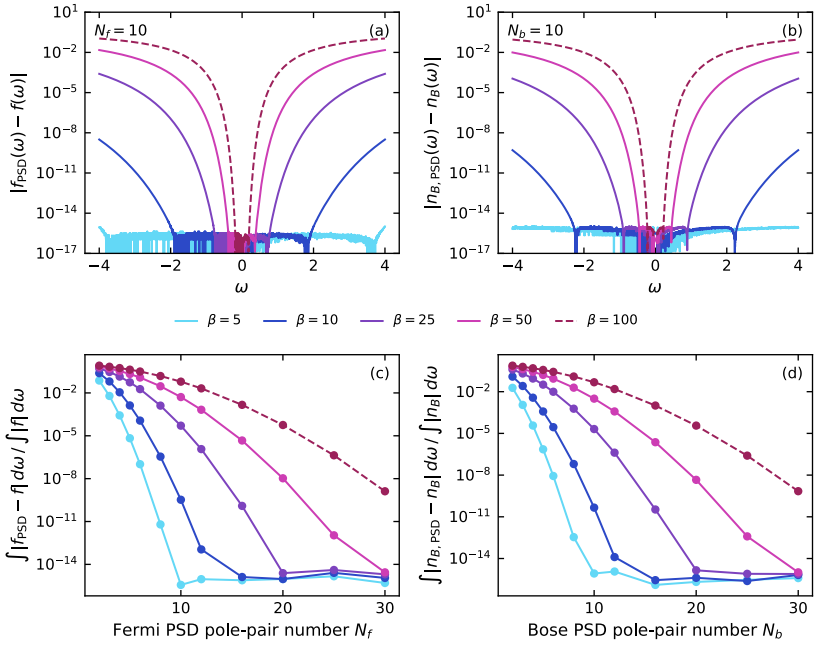
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How to perform real-axis integrals

- All Keldysh diagrams are integrals over the real axis of a distribution function times products of Green's functions. They all become sums over a finite (small) number of poles times a distribution function.
- Need to evaluate contractions of Green's functions, self-energies, and distribution functions. **Distribution functions** have an infinite number of poles
- Approximate the **distribution function** with a **pole approximation** within the region of support of the Green's function [Hu *et al.*, J. Chem. Phys. 133, 101106 (2010), Ozaki Phys. Rev. B 75, 035123 (2007)]
- Since there is **no integration grid**, there is no grid discretization to control and **no discretization error**.

Farid, Al-Eryani, Jażdżewska et al., in preparation (2026)

How to perform real-axis integrals



$$\Sigma_2^{>}(\omega)/U^2 = \int \frac{d\omega_1}{2\pi}\frac{d\omega_2}{2\pi} G^{>}(\omega_1) G^{>}(\omega_2) G^{<}(\omega_1+\omega_2-\omega) \,,$$
$$\Sigma_2^{<}(\omega)/U^2 = \int \frac{d\omega_1}{2\pi}\frac{d\omega_2}{2\pi} G^{<}(\omega_1) G^{<}(\omega_2) G^{>}(\omega_1+\omega_2-\omega) \,.$$

- Poles of the product are poles of the constituents. Multiply to get residues, then use residue theorem for integral

$$\Sigma_2^{>}(\omega)/U^2 = \left(\sum_k \sum_{j\in\mathrm{up}} \sum_{i\in\mathrm{up}} + \sum_j \sum_{i:(k-i)\in\mathrm{up}} \sum_{k\in\mathrm{up}} \right) \frac{A_i^{>} A_j^{>} A_k^{<}}{\omega - (\xi_i^{>} + \xi_j^{>} - \xi_k^{<})}$$
$$- \left(\sum_i \sum_{j\in\mathrm{up}} \sum_{k\in\mathrm{up}} + \sum_j \sum_{i:(k-i)\in\mathrm{up}} \sum_{i\in\mathrm{up}} \right) \frac{A_i^{>} A_j^{>} A_k^{<}}{\omega - (\xi_i^{>} + \xi_j^{>} - \xi_k^{<})} \,,$$

- Construct a Padé approximant of the distribution function accurate over the region of support of the spectral function

Farid, Al-Eryani, Jażdżewska et al., in prep. (2026)

How to solve geometric series

- Need to evaluate Dyson equations for Green's functions, self-energies and polarizations. Requires solving a geometric series or inverting pole representations.
- How to do this without resorting to evaluation? How to get a pole representation for G given one for Σ and G_0 directly in coefficient space?
- Write the self-energies and G_0 as ratios of polynomials. Then the location of the residues of G corresponds to the zero of a higher-order polynomial.
- Those zeros can be extracted by solving a (generally **non-Hermitian**) **eigenvalue problem** (for real poles and residues: known in non-interacting systems as the **resolvent problem**).
- The corresponding residues follow from an explicit formula.
- Solution of a Dyson or Polarization equation simplifies to a **linear algebra problem**: matrix diagonalization of a matrix of the size (#poles of Σ)+(#poles of G_0)

Farid, Al-Eryani, Jażdżewska et al., in preparation (2026)

How to solve geometric series

$$g_0(z) = \sum_{l=1}^{M_0} \frac{a_l}{z - \xi_l}, \qquad \operatorname{Im} \xi_l \le 0, \quad a_l \in \mathbb{C},$$

$$\sigma(z) = \sum_{k=1}^{M_\sigma} \frac{b_k}{z - \zeta_k}, \qquad \operatorname{Im} \zeta_k \le 0, \quad b_k \in \mathbb{C}.$$

$$g(z) = \frac{1}{g_0(z)^{-1} - \sigma(z)}.$$

Goal: Express g in terms of poles eta and residues c, directly from (a_j,xi_j) and (b_k, zeta_k)

$$D_0(z) := \prod_{l=1}^{M_0} (z - \xi_l), \qquad N_0(z) := \sum_{l=1}^{M_0} a_l \prod_{j \neq l} (z - \xi_j), \quad g_0(z) = N_0(z)/D_0(z),$$

$$D_\sigma(z) := \prod_{k=1}^{M_\sigma} (z - \zeta_k), \qquad N_\sigma(z) := \sum_{k=1}^{M_\sigma} b_k \prod_{j \neq k} (z - \zeta_j), \quad \sigma(z) = N_\sigma(z)/D_\sigma(z)$$

$$g(z) = \frac{N_0(z)\, D_\sigma(z)}{\mathcal{D}(z)} \qquad \mathcal{D}(z) := D_0(z)\, D_\sigma(z) - N_0(z)\, N_\sigma(z)$$

$$\deg D_0 = M_0$$

$$\deg D_\sigma = M_\sigma$$

Thus the poles eta of g are the zeros of D, and the residues cj are

$$c_j = \frac{N_0(\eta_j) D_\sigma(\eta_j)}{\mathcal{D}'(\eta_j)}$$

Turned the inversion into a problem of finding zeros of a polynomial of rank M0+Msigma.
Numerically this would be annoying (and expensive) but there is a simple trick: build

$$\mathcal{H} = \begin{pmatrix} \Xi & \mathbf{p}\mathbf{q}^T \\ \mathbf{r}\mathbf{s}^T & Z \end{pmatrix} \qquad \Xi = \operatorname{diag}(\xi_1, \ldots, \xi_{M_0}) \qquad p_l\, s_l = a_l \ (l = 1, \ldots, M_0)$$

$$Z = \operatorname{diag}(\zeta_1, \ldots, \zeta_{M_\sigma}) \qquad q_k\, r_k = b_k \ (k = 1, \ldots, M_\sigma)$$

Turned the inversion into a problem of finding zeros of a polynomial of rank M0+Msigma.
Numerically this would be annoying (and expensive) but there is a simple trick: build

$$\det(zI - \mathcal{H}) = \det(zI - \Xi)\, \det\big[(zI - Z) - \mathbf{r}\mathbf{s}^T (zI - \Xi)^{-1} \mathbf{p}\mathbf{q}^T\big]$$

$$\mathbf{s}^T (zI - \Xi)^{-1} \mathbf{p} = \textstyle\sum_l p_l s_l / (z - \xi_l) = g_0(z)$$

$$\det(A - \mathbf{r}\mathbf{q}^T \alpha) = (1 - \alpha\, \mathbf{q}^T A^{-1} \mathbf{r}) \det A$$

$$A = zI - Z,\ \alpha = g_0(z)$$

$$\det[(zI - Z) - g_0(z)\, \mathbf{r}\mathbf{q}^T] = D_\sigma(z)(1 - g_0(z)\, \sigma(z))$$

$$g_0 \sigma = N_0 N_\sigma / (D_0 D_\sigma)$$

$$\det(zI - \mathcal{H}) = D_0(z)\, D_\sigma(z)(1 - g_0(z)\, \sigma(z)) = D_0(z)\, D_\sigma(z) - N_0(z)\, N_\sigma(z) = \mathcal{D}(z)$$

Thus find the eigenvalues of H and those are the zeros of g. Then simply evaluate cj

Literature: Löwdin *J. Chem. Phys.* 19, 1396–1401 (1951), Löwdin, P.-O. Int. J. Quantum Chem., 21: 69-92 (1982), H. Feshbach, *Annals of Physics* **5**, 357–390 (1958), Brand, Annals of Physics **245**, 147–208 (1996); See also Chiarotti *et al.*, Physical Review Research **6**, L032023 (2024)

Farid, Al-Eryani, Jażdżewska et al., in preparation (2026)

How to avoid proliferation of poles – re-compression step

- **Many terms contribute** to a self-energy (spin sum, momentum sum, orbital sum, …). Each term results in a set of independent poles and residues. The resulting Self-energy has many more poles than the original Green's function; the Green's function after a Dyson equation more than the self-energy, …
- **Self-consistency** leads to a **proliferation of poles** that makes the problem intractable.
- **Recompression step**: We are not interested in the pole locations but in the spectral function. The spectral function is uniquely characterized by the **moments**.

- Real axis moments diverge so work with the **trigonometric moment problem**

Dong, Gull, arXiv:2607.11190

How to avoid proliferation of poles – re-compression step

- Use the moment representation – map real axis to unit circle
- **Compute trigonometric moments** using residue theorem as a pole sum for large M
- Ask the mathematical question: what is the **minimal number of poles** that can approximate the trigonometric moments to within a given precision?
- This is a so-called **truncated moment problem** solved by a **Prony approximation problem.**
- Solve the Prony problem with standard signal processing techniques (**Esprit**, MatrixPencil, …)
  Mathematically: SVD followed by least square fit.

- Given A, xi compute h

$$h_k = \sum_j^M A_j \xi_j^k$$

- Prony: given h, find A, xi

$$h_k \sim \sum_j^{\tilde{M}} \tilde{A}_j \tilde{\xi}_j^k$$

Phys. Rev. B 110, 035154 (2024)
Dong, Gull, arXiv:2607.11190

How to construct pole representations

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Solving the Prony problem: application of signal processing

- First exponential approximation and approximate extraction of poles rely on the solution of Prony problems. Here: ESPRIT.

$$\left| G_k - \sum_{i=0}^{K-1} w_i \gamma_i^k \right| \leq \varepsilon \text{ for all } 0 \leq k \leq 2N$$

- **Acoustics problem**: G_k: data sampled by a microphone. Wi: intensity. ɣ: frequency and phase [think exp(i omega t+phi)]
- **Radar problem**: G_k: data sampled by a phased array. Wi: radar crossection. ɣ: direction of arrival [think cos(theta)]
- We do not need the exact solution but the **minimum number of poles** that can approximate to within the precision epsilon

Phys. Rev. B 110, 235131 (2024)

Solving the Prony problem: application of signal processing

Goal: find residues w_i and poles gamma_i such that

$$G_k \approx \sum_{i=1}^{K} w_i \gamma_i^k.$$

Start by building structured matrix

$$H_{pq} = G_{p+q},$$

$$L \approx N/2,$$

$$q = 0, \ldots, N - L - 1.$$

$$p = 0, \ldots, L - 1,$$

Decompose H and set K to the point where the singular value is below tolerance

$$H = U \Sigma V^*.$$

Define shift matrices. Delete first and last row, respectively:

$$U_K = U(:, 1:K).$$

$$U_\uparrow = U_K(1 : L-1, :),$$
$$U_\downarrow = U_K(2 : L, :).$$

Compute Ψ and diagonalize it (size K)

$$\Psi = U_\uparrow^\dagger U_\downarrow.$$

$$\gamma_i = \lambda_i(\Psi), \quad i = 1, \ldots, K.$$

The eigenvalues ɣ of Ψ correspond to the poles.

Build the Vandermonde matrix V

$$V_{ki} = \gamma_i^k$$

Obtain the complex residues by approximating the data with a linear least square fit

$$w = \arg\min_w \|Vw - G\|_2,$$

Validate the residues, adjust K if needed.

$$r_k = G_k - \sum_{i=1}^{K} w_i \gamma_i^k.$$

Mathematically expensive step: SVD of moment matrix H. Number of moments needed is approximately the number of poles.

Moments are additive, Moment matrix is positive semidefinite for distribution functions like the spectral function.

Time extrapolation of real-time data, see Erpenbeck et al. *Phys. Rev. B 113, 115129 (2026)*

PRB 110, 235131 (2024)

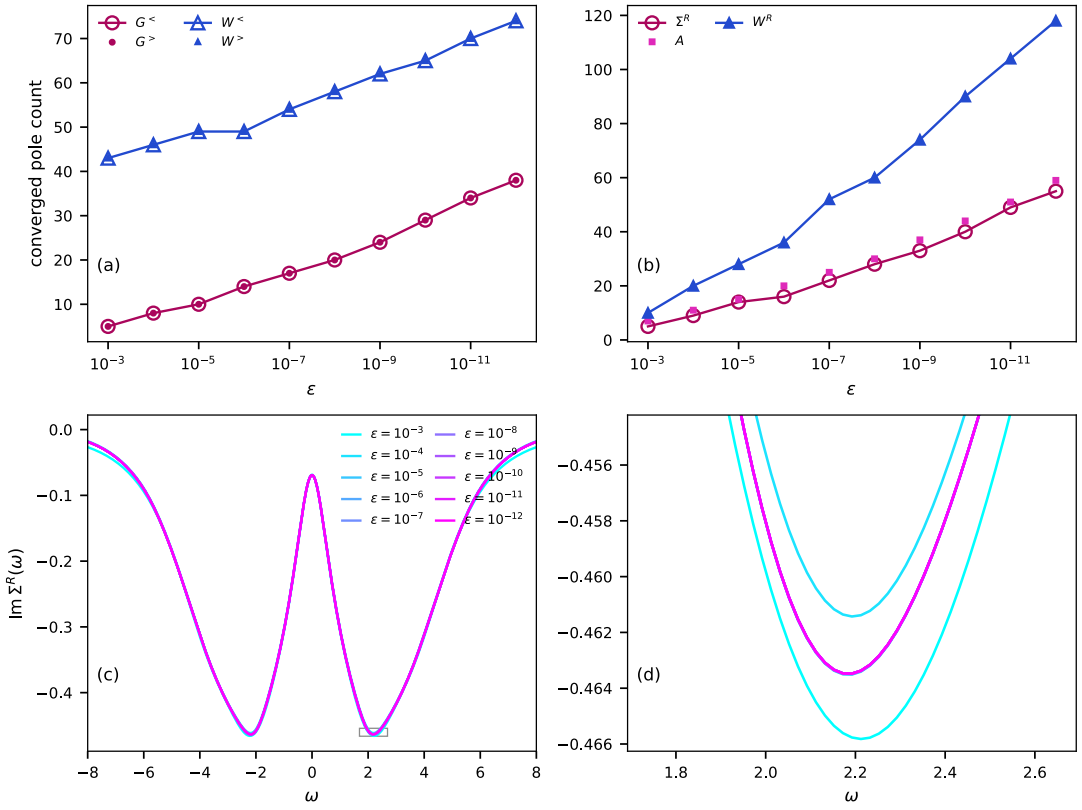
Some examples and applications

weak coupling diagrams uniform electron gas

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Farid, Al-Eryani, Jażdżewska et al., in preparation (2026)

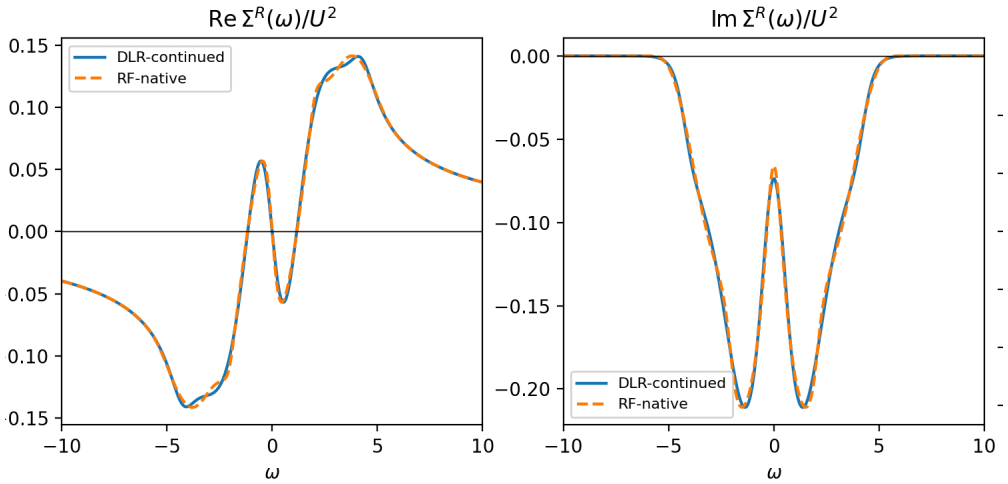
Convergence with poles M



- Adjust number of poles to reach a fixed error level
- Number of poles needed will depend on physics.
- For more than 4 poles of G, 40 of W, results look converged visually
- Errors can be made arbitrarily small by increasing the number of poles

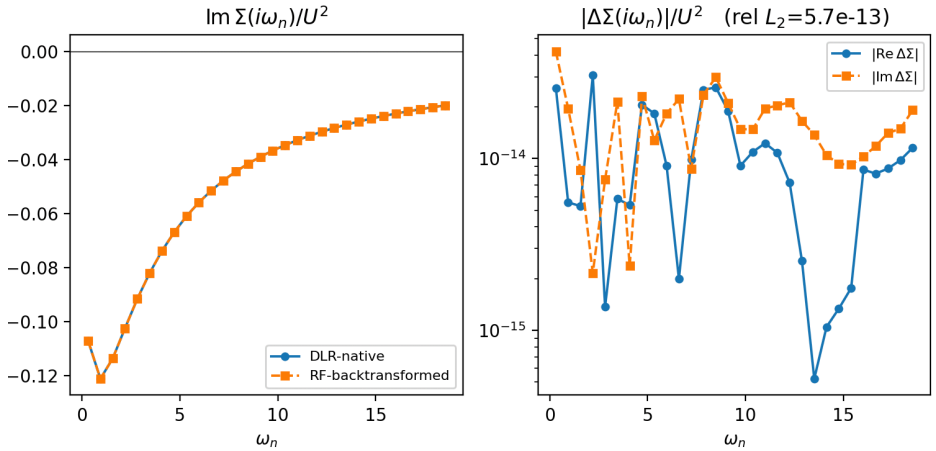
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Validation

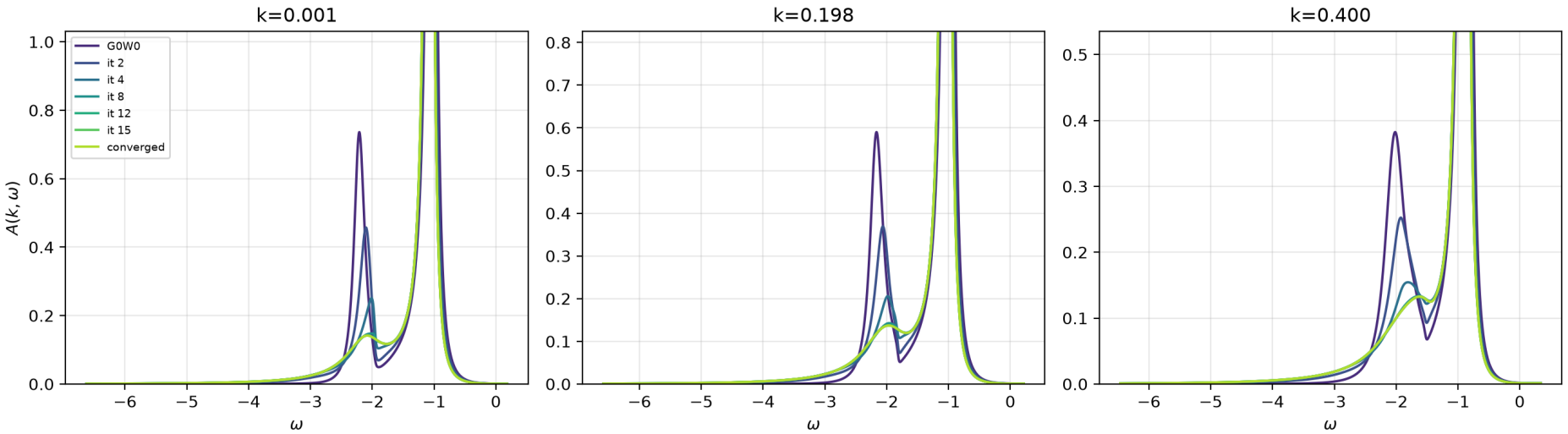


- Validation of Matsubara self-energies against back-continued spectra
- Expect agreement to numerical precision

- Validation of spectra against analytically continued Matsubara data
- Ill-conditioned A.C. regularized by MiniPole.



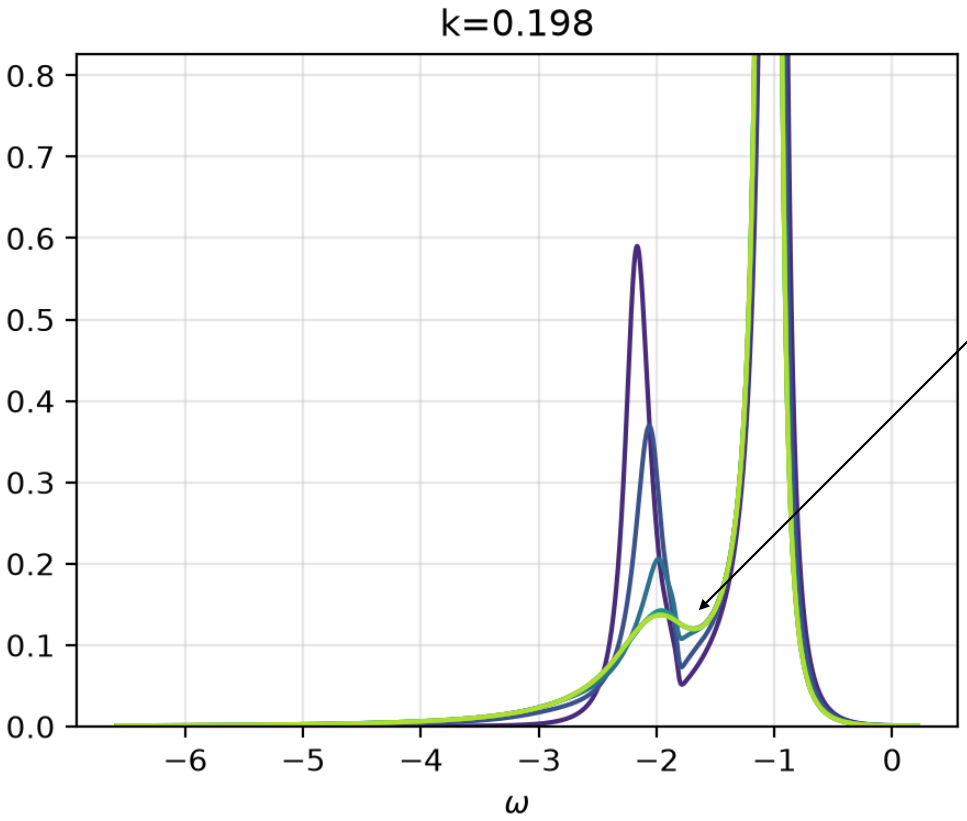
Application: UEG



- Paradigmatic correlation problem: the uniform electron gas. Here solved within G_0W_0 and scGW.
- Spectral function converged to very high precision (10^{-12})
- No analytical continuation, no grid discretization in frequency space, k-space dispersion resolved by Chebyshev interpolation of the moments

Farid, Al-Eryani, Jażdżewska et al., in preparation (2026)

Application: UEG



- This is the so-called '**satellite**'. It is weaker in sc-GW than in G_0W_0 but it does not disappear.

- Disappearance of satellites identified as a **weakness of self-consistent methods** and **justification for non-selfconsistency**.

Farid, Al-Eryani, Jażdżewska et al., in preparation (2026)

Application: Analytic Continuation

'Ill conditioned': Many complex functions approximate Matsubara data within its precision.

Regularization required. Typically: choose *smoothness.* (MaxEnt, SOM, ML, …)

Better regularization: Choose minimal information principle. Allows for sharp features as long as they carry low information

Reason for 'failure' of AC in standard methods is unsuitable regularization criterion

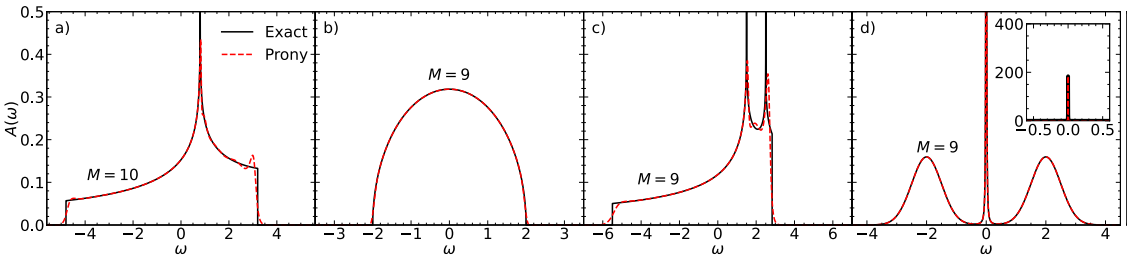


FIG. 3. Continuation of continuous spectral functions. From left to right: tight binding density of states of 2d square lattice with nearest- and next-nearest neighbor hopping. Semicircular density of states. Tight-binding density of states of the anisotropic triangular lattice. 'Kondo'-like spectral function. Shown are the exact input $A(\omega)$ in black, a continuation with Maximum Entropy (blue), SOM (purple), and a Prony fit (this method) in red. Maximum Entropy parameters fine-tuned to yield best spectra possible.

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Obtaining the Pole Approximation from Matsubara data

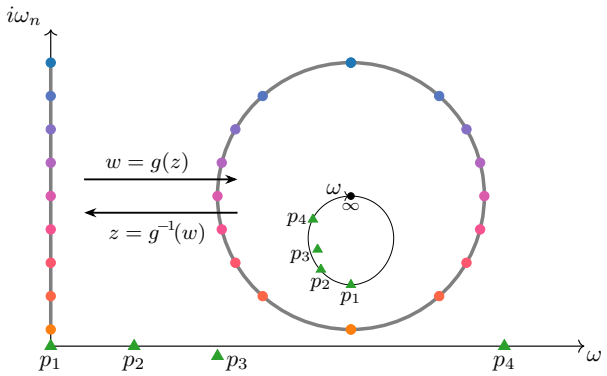
- Given Matsubara points, construct a smooth approximant on the Matsubara axis.

$$\left| G_k - \sum_{i=0}^{K-1} w_i \gamma_i^k \right| \leq \varepsilon \text{ for all } 0 \leq k \leq 2N$$

- This exponential fit minimizes ‘wiggles’, poles far away from data. Known as a **Prony problem**

- Joukowsky transform of line to unit circle

$$\begin{cases} w = g(z) & = z_{\mathrm{s}} - \sqrt{z_{\mathrm{s}}^2 + 1} \text{ with } z_{\mathrm{s}} = \frac{z - i\omega_{\mathrm{m}}}{\Delta\omega_{\mathrm{h}}} \\ z = g^{-1}(w) & = \frac{\Delta\omega_{\mathrm{h}}}{2}\left(w - \frac{1}{w}\right) + i\omega_{\mathrm{m}} \end{cases}$$



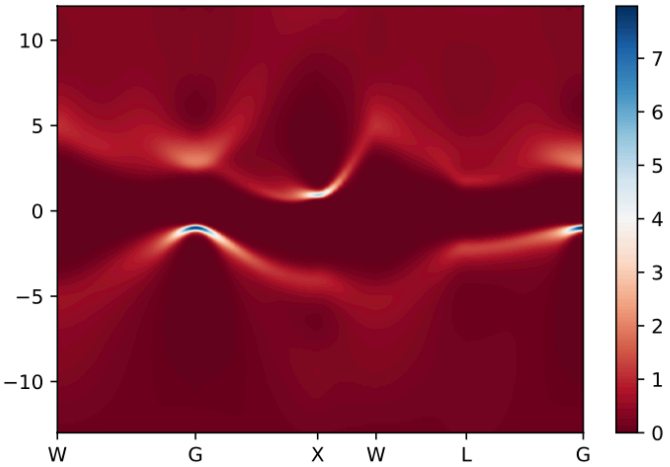
- Joukowsky transform maps the real axis into an ‘egg-shaped region’ in the interior of the unit circle. Maps poles close to zero.

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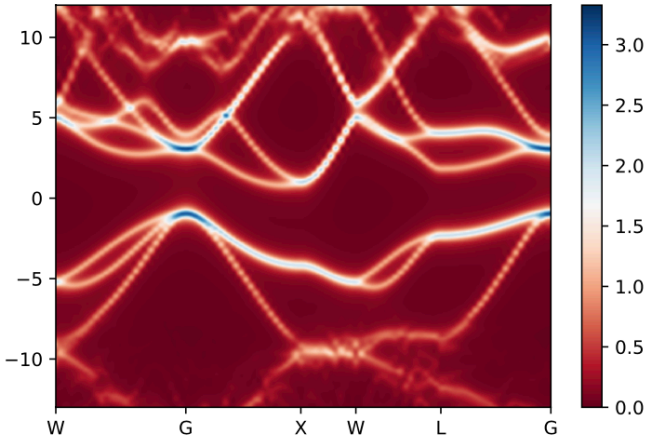
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Analytic Continuation Progress

Smoothness regularization: Maximum Entropy



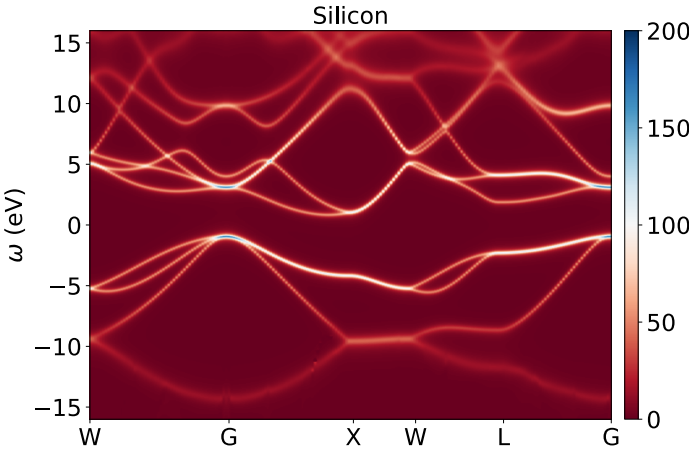
Maxent, orbital- and k-resolved



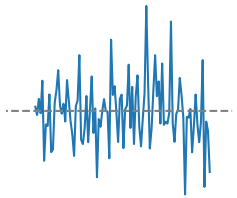
Nevanlinna, orbital- and k-resolved

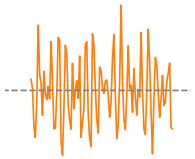
Causal Interpolation: Nevanlinna theory

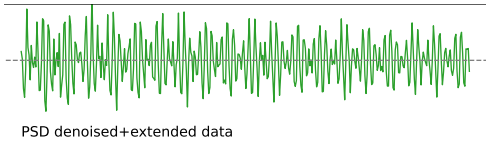
Minimal Information regularization: Prony/Esprit fit



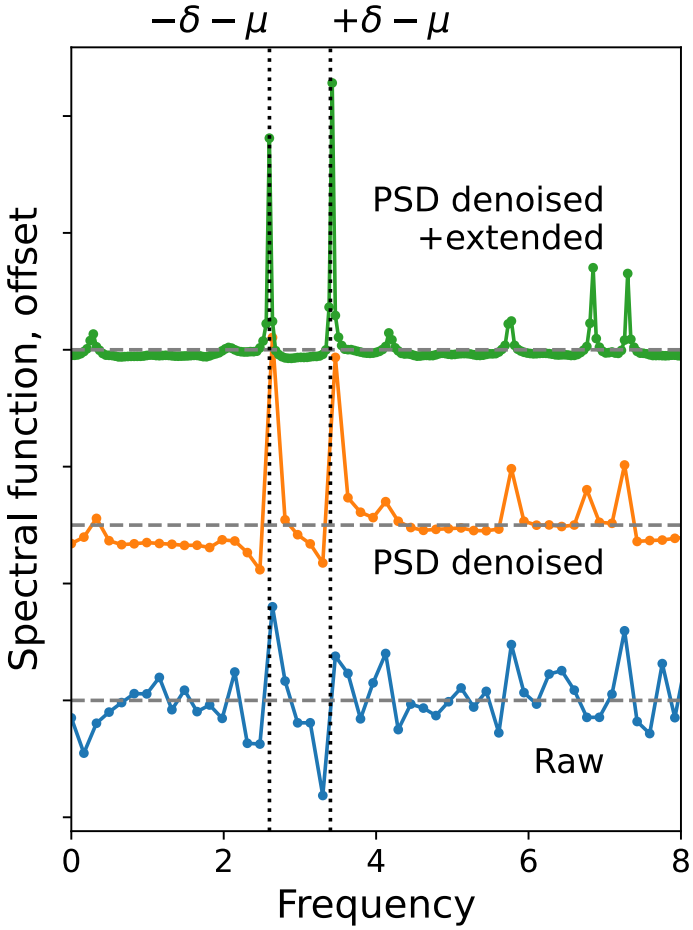
Causal Extension Problem







Very noisy real-time data. Use causality to project onto causal subspace. Then extend for long times (here using iteration, not Esprit)



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What can we do with this?

- **Numerical Analytic Continuation**: *Phys. Rev. B 110, 235131 (2024)*
  - Obtain the minimal number of poles needed to describe a Matsubara Green's function to within a given precision level
- "**Fitting" hybridization functions** with real poles: *Phys. Rev. B 110, 235131 (2024)*
  - Obtain the best set of N real frequency poles that approximate the Matsubara Green's function
- **Extending / denoising real-time data**: *Physical Review B 113 (11), 115129 (2026),*
  - Given is data up to some finite time. Find causal data for all times consistent with the input data to within some tolerance.
- **Representing spectral densities**: *J. Chem. Phys. 162, 214111 (2025)*
- **Fast calculations**: *Phys. Rev. B 110, 075158 (2024)*
  - Poles are incredibly easy to handle analytically. Precompute convolutions, perform math analytically

Collaborators



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Phys. Rev. B 110, 035154 (2024)

Thanks!

PhD and Postdoc positions available at the University of Warsaw, Poland, as part of the ERC Advanced project Quantum Algorithms





MIT licensed open source implementations

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